Modular arithmetic

1 Congruence
2 solution of linear equation
3 Fermais little theorem
4 Euler's theorem

I Congruence

If a & b are two integers then a is said to be congruence of b with modulo n.

if and only if when n divides a-b

it is denoted by

$$a \equiv b \pmod{n}$$

$$n \mid a-b$$

eg $7 \equiv 2 \pmod{5}$

$\frac{7-2}{5}$ = Quotient = 1 remainder = 0

eg $7 \equiv 3 \pmod{5}$

$\frac{7-3}{5}$ = Remainter $\neq 0$

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gcd - greatest common devisor
↓
hcf - highest common factor

(3, 15) → (HCF / GCD)

$$\begin{array}{r|l} 3 & 3 \\ \hline & 1 \end{array} \qquad \begin{array}{r|l} 3 & 15 \\ 5 & 5 \\ & 1 \end{array}$$

HCF = 3 = gcd
LCM = 3, 5

Type solution of linear equation / congruence

$ax \equiv b \pmod{n}$

Ex 1. Solve the linear congruence $3x \equiv 2 \pmod 7$

Solution :- $3x \equiv 2 \pmod 7$
$x = \{0, 1, 2, 3, 4, 5, 6\}$ [start from to mod-1 = 7-1 = 6]

$x = 0$, $0 \not\equiv 2 \pmod 7$ {atleast one solution max n-1 6}
$x = 1$, $3 \not\equiv 2 \pmod 7$
$x = 2$, $6 \not\equiv 2 \pmod 7$
$x = 3$, $9 \equiv 2 \pmod 7$
$x = 4$, $12 \not\equiv 2 \pmod 7$
$x = 5$, $15 \not\equiv 2 \pmod 7$
$x = 6$, $18 \not\equiv 2 \pmod 7$
$x = 3$ is the solution

option 1 $\frac{a-b}{n}$ [remainder should be 0]

option 2 $\frac{a}{n} = b$ then congruent.

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Ex 2 Solve the linear congruence
$6x = 3 \pmod 9$

solution $6x = 3 \pmod 9$
$x = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

$x=0 \quad 0 \neq 3 \pmod 9$
$x=1 \quad 6 \neq 3 \pmod 9$
$x=2 \quad 12 = 3 \pmod 9$
$x=3 \quad 18 \neq 3 \pmod 9$
$x=4 \quad 24 \neq 3 \pmod 9$
$x=5 \quad 30 = 3 \pmod 9$
$x=6 \quad 36 \neq 3 \pmod 9$
$x=7 \quad 42 \neq 3 \pmod 9$
$x=8 \quad 48 = 3 \pmod 9$

$$\begin{array}{r} 2 \\ 9\overline{)24} \\ -18 \end{array}$$

sir method

$2 \cdot 3\, x = 3 \cdot 1 \pmod{3 \cdot 3}$
$2x = 1 \pmod 3$

$x = \{0, 1, 2\}$
$x=0 \quad 0 \neq 1 \pmod 3$
$x=1 \quad 2 \neq 1 \pmod 3$
$x=2 \quad 4 = 1 \pmod 3$

$\therefore$ $x = 2$ is the solution

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Type : Fermat's little theorem

i $a^{p-1} = 1 \pmod p$

$2^{99} = 1 \pmod{100}$

ii If $ax \equiv b \pmod p$

$x = a^{p-2} \times b \pmod p$

Q1 show that $5^{38} \equiv 4 \pmod{11}$ by F.L.T

solution : $a^{p-1} \equiv 1 \pmod p$

$5^{10} \equiv 1 \pmod{11}$

$5^{38} \equiv 5^{30+8} \pmod{11}$

$\equiv 5^{30} \cdot 5^{8} \pmod{11}$

$\equiv (5^{10})^3 \cdot 5^{8} \pmod{11}$

$\equiv (1)^3 \cdot 5^{8} \pmod{11}$

$5^{38} \equiv 5^{8} \pmod{11}$

$5^2 \equiv 25 \equiv 3 \pmod{11}$

$5^3 = 125 \equiv 4 \pmod{11}$

jisk a remainc chota hai tak it

$5^{38} \equiv (5^2)^4 \pmod{11}$

$\equiv 3^4 \pmod{11}$

$\equiv 81 \pmod{11}$

$5^{38} = 4 \pmod{11}$

$$\begin{array}{r} 7 \\ 11\overline{)81} \\ = 77 \\ \hline 4 \end{array}$$

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Q show that $8^{57} = 3 \pmod 5$ by f.e.L.T

$$a^{p-1} = 1 \pmod p$$

$$8^4 = 1 \pmod 5$$

$\cancel{(8^4)^{14}} \equiv$

$$8^{57} \equiv 8^{56+1} \pmod 5$$
$$= 8^{56} \cdot 8^1 \bmod 5$$
$$= (8^4)^{14} \cdot 8^1 \pmod 5$$
$$= (1)^{14} \cdot 8^1 \pmod 5$$
$$= 8^1 \pmod 5$$
$$8^{57} \equiv 3 \pmod 5$$

$$5 \overline{)8}^{\,1} \quad -5 \quad 3$$

Q show that $3^{1000} = 4 \pmod 7$

$$3^6 \equiv 1 \pmod 7$$

divisor) — divident

$$3^{1000} \equiv 3^{996} \times 3^4 \pmod 7$$
$$\equiv ((3^6)^{166}) \times 3^4 \pmod 7$$

divident = Q x devisor + Rem $\equiv (1)^{166} \times 3^4 \pmod 7$

$$\equiv 3^4 \pmod 7$$

$$3^2 \equiv 9 \equiv 2 \bmod 7$$
$$3^2 = 27 \equiv 6 \bmod 7$$

$$7 \overline{)9}^{\,1} \quad -7 \quad 2 \qquad 7 \overline{)27}^{\,3} \quad -21 \quad 6$$

$$\equiv (3^2)^2 \bmod 7$$
$$\equiv (2)^2 \bmod 7$$
$$\equiv 4 \bmod 7$$
$$3^{1000} \equiv 4 \pmod 7$$

$$7 \overline{)4}^{\,0} \quad -0 \quad 4$$

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Ex4 ~~show that~~

Find least positive Residue (Remainder) of $2^{1000000}$ after dividing by 17 by F·L·T

$$2^{1000000} = \quad (\text{mod } 17)$$

By FLT

since $a^{p-1} \equiv 1 \pmod{p}$

$$2^{16} = 1 \pmod{17}$$

$$2^{1000000} \equiv (2^{16})^{62500} \pmod{17}$$

$$2^{1000000} \equiv (1)^{62500} \pmod{17}$$

$$2^{1000000} \equiv 1 \pmod{17}$$

Divident = Q × divisor + R

$$1000000 = 62500 \times 16 + ?$$

Ex5 solve $7x \equiv 12 \pmod{17}$

$$ax \equiv b \pmod{p}$$

~~7x =~~ by F·L·T

$$x = a^{p-2}\, b \pmod{p}$$

$$7x = 12 \pmod{17}$$

$$x = 7^{17-2} \times 12 \pmod{p}$$

$$= 7^{15} \times 12 \pmod{17} - 1$$

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$7^2 \equiv 49 \equiv 15 \pmod{17}$

$7^3 \equiv 343 \equiv 3 \pmod{17}$

$$\begin{array}{r} 20 \\ 17\overline{)343} \\ -340 \\ \hline 3 \end{array}$$

in calcu $- \underline{20 \cdot 73}$

$x \equiv (7^3)^5 \times 12 \pmod{17}$

$3^5 \times 12$

$\cancel{\equiv 2916 \pmod{17}}$

$$\begin{array}{r} 14 \\ 17\overline{)243} \\ -238 \\ \hline 5 \end{array}$$

$$\begin{array}{r} 3 \\ 17\overline{)60} \\ -51 \\ \hline 9 \end{array}$$

$x \equiv 5 \times 12 \pmod{17}$

$\equiv 60 \pmod{17}$

$x \equiv 9 \pmod{17}$

$x = 9$ is solution

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$4x = 11 \pmod{19}$

$x = 4^{17} \times 11 \pmod{19}$

$4^2 = 16 \equiv$

$4^3 = 64 \equiv 7 \pmod{19}$

$4^4 = 256 = 9$

$$\begin{array}{r} 3 \\ 19\overline{)64} \\ -57 \\ \hline 7 \end{array} \qquad \begin{array}{r} 13 \\ 19\overline{)256} \\ -247 \\ \hline 9 \end{array}$$

$x = \cancel{(4^3)^5 \neq 1}\ 4^{15} * 4^2 \times 11$

$(4^3)^5 * 4^2 \times 11 \pmod{19}$

$= 7^5 \times 176 \pmod{19}$

$$\begin{array}{r} 884 \\ 19\overline{)16807} \\ -16796 \\ \hline 11 \end{array}$$

$= 11 \times 176 \pmod{19}$

$= 1936 \pmod{19}$

$x = 17 \pmod{19}$

$$\begin{array}{r} 101 \\ 19\overline{)1936} \\ -1919 \\ \hline 17 \end{array}$$

$x = 17$ is the solution

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Euler's theorem

i] $a^{\phi(m)} \equiv 1 \pmod{m}$

ii] $ax = b \pmod{m}$

$$x \equiv a^{\phi(m)-1} \times b \pmod{m}$$

iii] $\phi(m) = m-1$

iv] $\phi(m \times n) = \phi(m) \times \phi(n)$

v] $\phi(m^2) = m^2 - m$

Ex] $\phi(2^2) = 2^2 - 2$
$= 2$

Q1] Solve the equation $7x \equiv 2 \pmod{15}$ by euler's theorem

$ax = b \pmod{m}$

by euler's theorm

$x = a^{\phi(m)-1} \times b \pmod{m}$

$7x = 2 \pmod{15}$

$a = 7 \quad b = 2 \quad m = 15$

$x = 7^{\phi(15)-1} \times 2 \pmod{15}$

$x = 7^{\phi(15)-1} \times 2 \pmod{15}$

$\phi(15) = \phi(3 \times 5) = \phi(3) \times \phi(5)$

$\phi(m) = m - 1$

$= (3-1) \times (5-1)$

$= 8$

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$$x = 7^{8-1} \times 2 \pmod{15}$$
$$= 7^7 \times 2 \pmod{15}$$
$$= 13 \times 2 \pmod{15}$$
$$= 26 \pmod{15}$$
$$= 11 \pmod{15}$$
$$= 11$$

$$\begin{array}{r} 54902 \\ 15\overline{)823543} \\ -\ 823530 \\ \hline 13 \end{array}$$

$$\begin{array}{r} 1 \\ 15\overline{)26} \\ -\ 15 \\ \hline 11 \end{array}$$

Q2] Solve the equation $9x = 2 \pmod{11}$ by Euler's method

$$x = 9^{(\phi(11)-1)} \times 2 \pmod{11}$$

$$\phi(11) = 11 - 1$$
$$= 10$$

$$\equiv 9^{(10-1)} \times 2 \pmod{11}$$
$$\equiv (9)^9 \times 2 \pmod{11}$$

$$9^2 \equiv 81 \equiv 4 \pmod{11}$$
$$9^3 = 729 \equiv 3 \pmod{11}$$

$$\begin{array}{r} 7 \\ 11\overline{)81} \\ -\ 77 \\ \hline 4 \end{array} \qquad \begin{array}{r} 66 \\ 11\overline{)729} \\ 726 \\ \hline 3 \end{array}$$

$$\equiv (9^3)^3 \times 2 \pmod{11}$$
$$\equiv (3)^3 \times 2 \pmod{11}$$
$$\equiv 27 \times 2 \bmod 11$$
$$\equiv 10 \bmod 11$$

$$\begin{array}{r} 4 \\ 11\overline{)54} \\ -\ 44 \\ \hline 10 \end{array}$$

$x = 10$ is the solution

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$x = 4$

Q) Solve $5x \equiv 3 \pmod{17}$
by euler's theorem

$$x = 5^{\phi(17)-1} \times 3 \mod 17$$

$$5^{(16-1)}$$

$$5^{(15)} \times 3$$

$$\times 3 \pmod{17}$$

$5^2 = 25 = \cancel{17}\,8 \pmod{17}$

$5^3 = 125 \equiv 6 \pmod{17}$

$$\begin{array}{r} 1 \\ 17\overline{)25} \\ -17 \\ \hline 8 \end{array}$$

$$\begin{array}{r} 7 \\ 17\overline{)125} \\ -119 \\ \hline 6 \end{array}$$

$(5^3)^5 \times 3 \pmod{17}$

$(6)^5 \times 3$

$23328 \mod 17$

$4 \pmod{17}$

$\underline{x = 4 \text{ is solution}}$

$$\begin{array}{r} 1372 \\ 17\overline{)23328} \\ -23324 \\ \hline 4 \end{array}$$

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Q] Use euler's theorem to compute
$7^{1615} \pmod{31}$

By euler's theorem

$$a^{\phi(m)} \equiv 1 \pmod{m}$$

$$7^{\phi(31)} \equiv 1 \pmod{31}$$

$$\phi(31) = 31 - 1 \qquad \phi(m) = m - 1$$
$$= 30$$

$$7^{30} \equiv 1 \pmod{31}$$

$$7^{1615} \equiv 7^{(30 \times 53) + 25} \pmod{31}$$

$$\begin{array}{r} 53 \\ 30 \overline{)\,1615} \\ -1590 \\ \hline 25 \end{array}$$

$$\equiv 7^{30} \times 7^{53} \cdot 7^{25} \pmod{31}$$
$$\equiv (7^{30})^{53} \cdot 7^{25} \pmod{31}$$
$$\equiv (1)^{53} \cdot (7)^{25} \pmod{31} \qquad 1615 = 30 \times 53 + 25$$
$$\equiv 7^{25} \pmod{31}$$

$$7^3 = 343 \equiv 2 \pmod{31}$$

$$= 7^{24+1} \pmod{31}$$
$$= 7^{24} \cdot 7^{1} \pmod{31}$$
$$= (7^3)^8 \cdot 7 \pmod{31}$$
$$= (2)^8 \cdot 7 \pmod{31}$$
$$1792 \bmod 31$$

$$\begin{array}{r} 57 \\ 31 \overline{)\,1792} \\ 1767 \\ \hline -25 \end{array}$$

$$7^{1615} = 25 \pmod{31}$$

$7^{1615} = 25 \pmod{31}$ is the solution

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